

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12388305
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Tanaka; Hiroaki

Compressor with stator having large diameter and small diameter parts and welding parts joining housing and stator

Abstract

A compressor includes: a compressor; and a motor that is disposed in the compressor housing. The motor has a rotor provided coaxially with respect to a rotary shaft of the compression part, and a stator disposed on an outer peripheral side of the rotor. In the stator, a plurality of large-diameter parts in contact with an inner peripheral surface of the compressor housing and a plurality of small-diameter parts, each of the small-diameter parts having a distance from a rotation center of the motor to an outer peripheral surface, the distance being smaller than the distance of the large-diameter part, are formed in a circumferential direction of the stator. The stator is provided with a plurality of welded parts joining the inner peripheral surface of the compressor housing with the outer peripheral surface of the stator, and the welded parts are provided in the small-diameter part of the stator.

Inventors:	Tanaka; Hiroaki (Kanagawa, JP)
Applicant:	FUJITSU GENERAL LIMITED (Kanagawa, JP)
Family ID:	81383705
Assignee:	FUJITSU GENERAL LIMITED (Kanagawa, JP)
Appl. No.:	18/032091
Filed (or PCT Filed):	September 22, 2021
PCT No.:	PCT/JP2021/034830
PCT Pub. No.:	WO2022/091655
PCT Pub. Date:	May 05, 2022

Prior Publication Data

Document Identifier

US 20230392589 A1

Publication Date

Dec. 07, 2023

Foreign Application Priority Data

JP

2020-182495

Oct. 30, 2020

Publication Classification

Int. Cl.: H02K1/18 (20060101); F04B35/04 (20060101); H02K7/14 (20060101)

U.S. Cl.:

CPC H02K1/185 (20130101); F04B35/04 (20130101); H02K7/14 (20130101);

Field of Classification Search

CPC: H02K (1/185); H02K (1/18); H02K (1/182); H02K (5/04); H02K (5/24); H02K (7/09); H02K (11/30); H02K (5/00); H02K (5/06); H02K (5/08); H02K (5/10); H02K (2213/03); F16C (32/047)

USPC: 310/89; 310/415; 310/431-433; 310/427; 310/216.127; 310/216.129; 310/88

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
493337	12/1892	Parshall	310/265	H02K 15/12
1703087	12/1928	Daun	310/216.118	H02K 1/185
1771475	12/1929	Wright	310/407	H02K 1/185
1795882	12/1930	Noden	310/216.049	H02K 1/185
1822096	12/1930	Hollander	310/426	H02K 1/185
2001799	12/1934	Seyfried	310/414	H02K 1/185
2011060	12/1934	Leland	310/216.135	H02K 1/185
2023259	12/1934	Anderson	310/426	H02K 1/185
2251674	12/1940	Gillen	310/216.049	H02K 17/10
2478316	12/1948	Potter	310/216.127	H02K 1/185
2523520	12/1949	Reinhard	310/410	H02K 1/185
2689394	12/1953	Bentjens	407/51	B23C 5/2295
2711492	12/1954	Ballman	164/109	H02K 1/185
2876371	12/1958	Wesolowski	310/216.049	H02K 1/16
2939021	12/1959	Gilchrist	310/91	H02K 1/185
2977491	12/1960	Hueffed	310/414	H02K 1/185
4227109	12/1979	Mulach	310/216.049	H02K 1/16
4286187	12/1980	Binder	310/90	H02K 7/1815
4603273	12/1985	McDonald	310/90	H02K 15/16
4642502	12/1986	Carpenter	310/216.013	H02K 1/2781

4765054	12/1987	Sauerwein	310/216.115	H02K 3/522
4797602	12/1988	West	310/156.28	H02K 1/278
4972113	12/1989	Newberg	310/410	H02K 15/16
4992686	12/1990	Heine	310/67R	H02K 5/225
5015904	12/1990	Kleemann	310/194	H02K 3/325
5319270	12/1993	Tanaka	310/216.096	H02K 11/21
5430338	12/1994	McMillan	310/91	H02K 5/15
5742991	12/1997	Kurth	425/589	B29C 45/1747
5767602	12/1997	Sargeant	310/58	H02K 5/207
5793136	12/1997	Redzic	310/112	H02K 16/02
5796190	12/1997	Takeda	310/156.31	H02K 1/32
5818131	12/1997	Zhang	310/90.5	F04D 13/0646
5821647	12/1997	Takehara	310/90	H02K 7/14
5850679	12/1997	Hoffman	29/252	B25B 27/026
5859486	12/1998	Nakahara	310/216.084	H02K 1/148
5949169	12/1998	Niimi	310/216.127	H02K 23/04
6040645	12/1999	Lynch	310/227	H02K 23/54
6124567	12/1999	Feldhausen	310/52	B23K 9/1006
6225722	12/2000	Rupp	310/91	H02K 21/22
6305989	12/2000	Quadir	439/685	H01R 13/533
6310320	12/2000	Kraus	219/133	B23K 9/1075
6317963	12/2000	Powers	310/58	H02K 15/14
6320287	12/2000	Watson	310/91	H02K 1/185
6346760	12/2001	Boardman, IV	310/216.118	H02K 1/16
6498417	12/2001	Fuller	310/431	H02K 1/185
6589018	12/2002	Chen	310/71	F04D 29/263
6713930	12/2003	Shah	310/216.076	H02K 1/185
6720699	12/2003	Shah	310/182	H02K 1/16
6737775	12/2003	Hatz	322/4	H02K 7/1815
6796021	12/2003	Fuller	310/91	H02K 15/00
6856064	12/2004	Masumoto	310/216.008	H02K 15/021
6866487	12/2004	Abe	417/410.5	F04C 18/0215
6933640	12/2004	Schurter	310/85	H02K 1/185
6960861	12/2004	Yoneda	310/216.012	H02K 1/148
6979930	12/2004	Harada	310/216.004	H02K 15/021
7397163	12/2007	Cook	310/427	H02K 1/187
7471000	12/2007	Ruiz	320/105	H02J 7/1415
7560838	12/2008	van der Woude	310/64	F02C 7/32
7583001	12/2008	Lu	310/67R	H02K 1/2791
7653986	12/2009	Majernik	29/598	H02K 15/028
7687952	12/2009	York	310/263	H02K 1/28
7687954	12/2009	Neet	310/263	H02K 19/24
7719157	12/2009	Yukitake	310/216.049	F04C 15/008
7827668	12/2009	McKee	29/515	E21B 17/20
7847444	12/2009	Kingman	310/91	H02K 5/203
7868509	12/2010	Yoshino	310/216.131	H02K 1/16
7919898	12/2010	Wang	310/216.127	H02K 1/276
7928617	12/2010	Tsukashima	310/51	H02K 7/20
7937823	12/2010	Alfermann	310/216.008	H02K 15/022
8138649	12/2011	Bradfield	310/156.19	H02K 21/044

8487502	12/2012	Kaiser	310/216.049	H02K 1/16
8587165	12/2012	Zahora	310/58	H02K 9/06
8643246	12/2013	Allen	310/216.029	H02K 1/18
8829743	12/2013	Watanabe	310/60A	H02K 9/19
8941282	12/2014	Allen	310/216.049	H02K 1/18
9136746	12/2014	Ikuta	N/A	H02K 15/066
9263921	12/2015	Tanavde	N/A	H02K 3/46
9287741	12/2015	Nobata	N/A	H02K 15/14
9496765	12/2015	Zheng	N/A	H02K 1/185
9509182	12/2015	Yamarthi	N/A	H02K 1/185
10075047	12/2017	Zhou	N/A	H02K 1/185
10177631	12/2018	Hopkins	N/A	H02K 5/15
10393082	12/2018	Ihde	N/A	H02J 7/1415
10483817	12/2018	Sakurai	N/A	H02K 1/28
10673289	12/2019	Nakagawa	N/A	H02K 1/148
10715006	12/2019	Bender	N/A	H02K 7/025
10840773	12/2019	Takano	N/A	H02K 9/227
10855134	12/2019	Ishii	N/A	H02K 7/083
10907627	12/2020	Ihde	N/A	F04B 17/03
11286956	12/2021	Kajikawa	N/A	H02K 7/14
11658525	12/2022	Coppola	310/54	H02K 15/028
2001/0038797	12/2000	Makino	417/410.3	H02K 7/14
2002/0060501	12/2001	Rafaelof	310/67R	H02K 5/24
2003/0184182	12/2002	Smith, Jr.	310/216.084	H02K 15/022
2003/0214197	12/2002	De Luca	310/216.084	H02K 1/148
2004/0217669	12/2003	Fujii	310/216.045	H02K 1/148
2004/0256932	12/2003	Kurosawa	310/89	H02K 1/185
2005/0034295	12/2004	Meacham	29/598	H02K 1/02
2005/0189837	12/2004	Lee	310/215	H02K 3/522
2006/0071574	12/2005	Stewart	310/216.132	H02K 1/146
2006/0279160	12/2005	Yoshinaga	310/216.049	H02K 1/148
2007/0182271	12/2006	Sugishima	310/216.012	H02K 15/095
2009/0085415	12/2008	Lonel	310/43	H02K 1/148
2009/0212654	12/2008	Kaneiwa	310/214	H02K 3/50
2009/0212655	12/2008	Miyairi	310/216.118	H02K 5/10
2009/0289522	12/2008	Buban	310/216.113	H02K 1/148
2010/0007236	12/2009	Sano	310/216.135	H02K 1/148
2010/0021321	12/2009	Koike	310/216.136	F04C 23/008
2010/0135830	12/2009	Yasuda	310/216.069	H02K 1/146
2010/0148620	12/2009	Ishizuka	310/201	H02K 3/12
2010/0270887	12/2009	Ishizuka	310/214	H02K 3/50
2010/0308687	12/2009	George	310/216.135	H02K 1/185
2011/0121680	12/2010	Boardman, IV	310/216.129	H02K 1/185
2012/0001515	12/2011	Kudose	310/216.009	H02K 1/16
2012/0139385	12/2011	Sawada	310/216.074	H02K 1/148
2012/0153749	12/2011	Chun	310/59	H02K 5/207
2012/0169173	12/2011	Jang	310/215	H02K 3/345
2012/0206009	12/2011	Jang	310/179	H02K 3/522
2013/0089428	12/2012	Hottier	416/223R	F01D 5/00
2013/0106252	12/2012	Yanagida	310/68B	H02K 11/225

2013/0140939	12/2012	Asaga	29/598	H02K 15/02
2013/0187517	12/2012	Asao	310/68D	H02K 5/04
2013/0207509	12/2012	Shimaguchi	310/216.113	H02K 1/185
2013/0313922	12/2012	Kim	310/44	H02K 3/522
2014/0167558	12/2013	Woo	310/216.113	H02K 5/24
2014/0197705	12/2013	Sato	310/91	H02K 1/185
2014/0197715	12/2013	Roopnarine	310/216.106	H02K 9/02
2014/0210284	12/2013	Banba	310/43	H02K 15/021
2014/0231483	12/2013	Takaichi	225/105	B26F 3/00
2014/0265683	12/2013	Hossain	310/89	H02K 1/185
2014/0333163	12/2013	Horii	310/59	H02K 9/10
2014/0354107	12/2013	Alfermann	310/216.113	H02K 15/14
2014/0360334	12/2013	Singer-Schnoeller	29/428	B23B 31/11
2015/0000114	12/2014	Matsushita	29/729	H02K 15/03
2015/0022051	12/2014	Meng	524/514	B29C 45/0001
2015/0069865	12/2014	Alfermann	310/54	H02K 9/19
2015/0076940	12/2014	Ohashi	310/58	H02K 5/04
2015/0143691	12/2014	Takaichi	29/738	H02K 15/03
2015/0222151	12/2014	Semken	29/452	H02K 1/28
2015/0364966	12/2014	Maihara	310/51	H02K 5/24
2016/0079817	12/2015	Hayashi	29/598	H02K 1/22
2016/0087493	12/2015	Kanakamedala	310/216.129	H02K 1/18
2016/0285324	12/2015	Carrasco	N/A	H02K 1/185
2016/0285342	12/2015	Carrasco	N/A	H02K 7/09
2017/0182587	12/2016	Tokoro	N/A	B23K 20/123
2017/0201135	12/2016	Kai	N/A	H02K 3/18
2017/0284396	12/2016	Fukasaku	N/A	F04C 21/00
2017/0313283	12/2016	Kojima	N/A	H02K 1/146
2018/0043420	12/2017	Handwerker	N/A	B21K 25/005
2018/0138762	12/2017	Jeong	N/A	H02K 5/203
2018/0248439	12/2017	McGrew, Jr.	N/A	H02K 1/185
2018/0351435	12/2017	Takano	N/A	H02K 9/223
2019/0173334	12/2018	Takahashi	N/A	H02K 19/22
2020/0144882	12/2019	Tang	N/A	H02K 1/185
2022/0094233	12/2021	Seo	N/A	H02K 5/225

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2006-191702	12/2005	JP	N/A
2010-081659	12/2009	JP	N/A
2011-019398	12/2010	JP	N/A
2011-050151	12/2010	JP	N/A
2011-055576	12/2010	JP	N/A
2012-161126	12/2011	JP	N/A
2017-034819	12/2016	JP	N/A

OTHER PUBLICATIONS

JP 2010081659A English Translation. cited by examiner
JP 2011019398a English Translation. cited by examiner
JP 2017034819a English Translation. cited by examiner

Background/Summary

CROSS REFERENCE TO PRIOR APPLICATION

(1) This application is a National Stage Patent Application of PCT International Patent Application No. PCT/JP2021/034830 (filed on Sep. 22, 2021) under 35 U.S.C. § 371, which claims priority to Japanese Patent Application No. 2020-182495 (filed on Oct. 30, 2020), which are all hereby incorporated by reference in their entirety.

FIELD

(2) The present invention relates to a compressor.

BACKGROUND

(3) For example, a rotary compressor is provided with a motor that is disposed in a compressor housing, and an outer peripheral surface of a stator of the motor is welded with and fixed to an inner peripheral surface of the compressor housing. In manufacturing processes of a rotary compressor of this type, it is known that a stator is temporarily joined with a compressor housing by shrink-fitting (A method of joining two parts by utilizing thermal expansion and contraction. [1] The inner diameter of a hole is enlarged by heating and expanding the part positioned in the outer side and provided with the hole. [2] The part positioned in the inner side is fitted into the hole of the part in the outer side with the enlarged diameter. [3] When the part in the outer side is cooled, the inner diameter of the hole is narrowed, and the two parts are mutually fixed since the part in the outer side tightens the part in the inner side.) the compressor housing onto the stator, and the compressor housing on which the stator is fitted is then conveyed to a welding process to carry out welding.

CITATION LIST

Patent Literature

(4) Patent Literature 1: JP 2006-191702 A Patent Literature 2: JP 2011-55576 A

SUMMARY

Technical Problem

(5) In the motor that is disposed in the compressor housing, when distortion occurs in the stator, the magnetic characteristics of the stator are deteriorated, iron loss increases, and motor efficiency is decreased. Therefore, when temporary joining of the stator is to be carried out by shrink-fitting of the compressor housing, generally, the inner diameter of the compressor housing is formed with respect to the outer diameter of the stator so as to implement transition-fitting (a state in which the outer diameter of the stator can be larger or smaller than the inner diameter of the compressor housing because of variations within the range of tolerance), which is between interference-fitting of fitting tolerance (a state in which the outer diameter of the stator is always larger than the inner diameter of the compressor housing within the range of tolerance) and clearance-fitting (a state in which the outer diameter of the stator is always smaller than the inner diameter of the compressor housing within the range of tolerance). By virtue of this, the generation of distortion in the stator caused by the external force applied from the shrink-fitted compressor housing, is suppressed. Then, after the stator is temporarily joined with the compressor housing by shrink-fitting, the stator is fixed to the compressor housing by welding. In other words, the stator is firmly fixed to the compressor housing while distortion of the stator is decreased, by fixing the stator in the compressor housing by welding, while the temporarily joined state of the stator disposed in the compressor housing is maintained by the transition-fitting.

(6) However, even in the case in which the inner diameter of the compressor housing is formed to implement the transition-fitting, if dimensional variations result in the vicinity of a lower limit of dimension tolerance, the interference (the difference between the outer diameter of the stator and the inner diameter of the compressor housing in the case in which the outer diameter of the stator is larger than the inner diameter of the compressor housing) may become large as well as the interference-fitting. If the interference becomes large, motor efficiency is lowered due to the distortion generated in the stator tightened by the compressor housing, and there is no longer the advantage of fixing the stator onto the compressor housing by welding.

(7) The disclosed techniques have been accomplished in view of the foregoing, and it is an object to provide a compressor which can decrease the distortion generated in a stator when the stator is fixed to a compressor housing.

Solution to Problem

(8) According to an aspect of an embodiments in the present application, a compressor includes: a compressor housing; a compression part that is disposed in the compressor housing and compresses a cooling medium; and a motor that is disposed in the compressor housing and drives the compression part, wherein the motor has a rotor provided coaxially with respect to a rotary shaft of the compression part, and a stator disposed on an outer peripheral side of the rotor, in the stator, a plurality of large-diameter parts in contact with an inner peripheral surface of the compressor housing and a plurality of small-diameter parts, each of the small-diameter parts having a distance from a rotation center of the motor to an outer peripheral surface, the distance being smaller than the distance of the large-diameter part, are formed in a circumferential direction of the stator, and the stator is provided with a plurality of welded parts joining the inner peripheral surface of the compressor housing with the outer peripheral surface of the stator, and the welded parts are provided in the small-diameter parts of the stator.

Advantageous Effects of Invention

(9) According to a mode of a compressor disclosed in the present application, the distortion generated in a stator when the stator is fixed to a compressor housing can be decreased, and decrease in motor efficiency can be suppressed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a vertical cross-sectional view illustrating a compressor of an embodiment.

(2) FIG. 2 is a transverse cross-sectional view illustrating a three-phase motor of the compressor of the embodiment.

(3) FIG. 3 is a plan view illustrating a stator of the three-phase motor in the embodiment.

(4) FIG. 4 is a plan view illustrating a stator core in the embodiment.

(5) FIG. 5 is a plan view illustrating a main part of the stator core in the embodiment.

(6) FIG. 6 is a plan view enlarging and illustrating a small-diameter part of the stator core in the embodiment.

(7) FIG. 7 is a plan view enlarging and illustrating a through-hole of the stator core in the embodiment.

(8) FIG. 8 is a plan view for describing the external force and the stress which act on the stator core in manufacturing processes of the compressor of the embodiment.

(9) FIG. 9 is a diagram illustrating deformation generated in the stator core in the embodiment.

(10) FIG. 10 is a diagram illustrating correction of the deformation of the stator core in the embodiment.

(11) FIG. 11 is a graph illustrating that deformation of the stator core in the embodiment is corrected by welding.

(12) FIG. 12 is a graph illustrating relations between interferences of the stator core in the embodiment and a comparative example and decrease rates of motor efficiency.

(13) FIG. 13 is a graph for describing APF efficiency in the embodiment.

DESCRIPTION OF EMBODIMENTS

(14) Hereinafter, an embodiment of a compressor disclosed in the present application will be described in detail based on drawings. Note that the compressor disclosed in the present application is not limited by the following embodiment.

Embodiment

(15) FIG. 1 is a vertical cross-sectional view illustrating a compressor of an embodiment. As illustrated in FIG. 1, a compressor 1 is a so-called rotary compressor and is provided with a container 2 serving as a compressor housing, a shaft 3 serving as a rotary shaft, a compression part 5, and a three-phase motor 6. The container 2 is formed of a metallic material and forms a sealed internal space 7. The internal space 7 is formed in an approximately columnar shape. The container 2 is formed so that, when the container is vertically placed on a horizontal plane, the central axis of the column forming the internal space 7, is parallel to the perpendicular direction. In the container 2, an oil reservoir 8 is formed in a lower part of the internal space 7. In the oil reservoir 8, refrigerant oil 8a, which is lubricating oil which lubricates the compression part 5, is stored. Intake tubes 11 serving as intake parts, which take in a cooling medium, and a discharge tube 12 serving as a discharge part, which discharges the compressed cooling medium, are connected to the container 2. The shaft 3 serving as the rotary shaft is formed in a rod shape and is disposed in the internal space 7 of the container 2 so that one end thereof is disposed in the oil reservoir 8. The shaft 3 is supported, by the container 2, rotatably about the central axis of the column which forms the internal space 7. The shaft 3 supplies the refrigerant oil 8a, which is stored in the oil reservoir 8, to the compression part 5 by rotating.

(16) The compression part 5 is disposed in the lower part of the internal space 7 and is disposed above the oil reservoir 8. The compressor 1 is further provided with an upper muffler cover 14 and a lower muffler cover 15. The upper muffler cover 14 is disposed in an upper part of the compression part 5 in the internal space 7. The upper muffler cover 14 forms an upper muffler chamber 16 therein. The lower muffler cover 15 is provided in a lower part of the compression part 5 in the internal space 7 and is disposed above the oil reservoir 8. The lower muffler cover 15 forms a lower muffler chamber 17 therein. The lower muffler chamber 17 is communicated with the upper muffler chamber 16 via a communication path (not illustrated) formed in the compression part 5. A compressed cooling-medium discharge hole 18 is formed between the upper muffler cover 14 and the shaft 3, and the upper muffler chamber 16 is communicated with the internal space 7 via the compressed cooling-medium discharge hole 18.

(17) When the shaft 3 rotates, the compression part 5 compresses the cooling medium, which is supplied from the intake tubes 11, and supplies the compressed cooling medium to the upper muffler chamber 16 and the lower muffler chamber 17. The cooling medium has compatibility with the refrigerant oil 8a. The three-phase motor 6 is disposed above the compression part 5 in the internal space 7.

(18) FIG. 2 is a transverse cross-sectional view illustrating the three-phase motor 6 of the compressor 1 of the embodiment. As illustrated in FIG. 1 and FIG. 2, the three-phase motor 6 is provided with a rotor 21 and a stator 22. The rotor 21 is formed in a columnar shape by stacking a plurality of metal plates and is integrated by later-described swaged parts 28 and a plurality of rivets 9. The shaft 3 is inserted in the center of the rotor 21, and the rotor 21 is fixed with respect to the shaft 3. In the rotor 21, six slit-shaped magnet embedding holes 10a are formed so as to form sides of a hexagon about the shaft 3. The magnet embedding holes 10a are formed at a predetermined interval in a circumferential direction of the rotor 21. In the magnet embedding holes 10a, plate-shaped permanent magnets 10b are embedded.

(19) The stator 22 is formed in an approximately cylindrical shape and is disposed so as to

surround an outer peripheral side of the rotor **21**. The stator **22** is provided with an annular stator core **23**, an upper insulator **24**, a lower insulator **25**, and a plurality of winding wires **46** (see FIG. **1**). The stator **22** and the container **2** are welded by a plurality of welded parts **20** that are formed in the circumferential direction of the stator **22**. In other words, the outer peripheral surface of the stator core **23** is spot-welded by the welded parts **20** with respect to an inner peripheral surface of the container **2**. The welded parts **20** are provided at intervals in the circumferential direction of the stator **22** (about the axis of the shaft **3**). Also, a plurality of welded parts **20** is provided also in the axial direction of the stator **22** (axial direction of the shaft **3**).

(20) The upper insulator **24** is fixed to an upper end of the stator core **23**. The lower insulator **25** is fixed to a lower end of the stator core **23**. Also, as illustrated in FIG. **1** and FIG. **2**, in the stator core **23**, an insulative film **26** is inserted along the inner peripheral surfaces of slots between later-described stator-core teeth parts **32-1** to **32-9**, and the stator core **23** and the winding wires **46** are insulated from each other by the insulative film **26**. The insulative film **26** is formed of a resin material such as polyethylene terephthalate (PET) or polyethylene naphthalate (PEN). Also, the upper insulator **24** and the lower insulator **25** are formed of resin materials, and are insulative members which insulate the stator core **23** and the winding wires **46** from each other.

Structure of Stator

(21) FIG. **3** is a plan view illustrating the stator **22** of the three-phase motor **6** in the embodiment. As illustrated in FIG. **3**, the winding wires **46**, which are electric wires, are wound around each of the later-described plurality of stator-core teeth parts **32-1** to **32-9** in the stator core **23**. In each of the stator-core teeth parts **32-1** to **32-9**, a wound part **45** is formed by each winding wire **46**. The three-phase motor **6** in the embodiment is a motor of a concentrated winding type with six poles and nine slots (see FIG. **2**). The plurality of winding wires **46** is provided with a plurality of U-phase winding wires **46-U1** to **46-U3**, a plurality of V-phase winding wires **46-V1** to **46-V3**, and a plurality of W-phase winding wires **46-W1** to **46-W3**. Also, in the stator **22**, neutral wires, which are extended from the wound parts **45** and gathered in a bundle, are covered with insulative tubes and inserted in the gaps between the wound parts **45**, which are mutually adjacent in the circumferential direction of the stator **22** (rotation direction of the rotor **21**) (see FIG. **2**).

(22) FIG. **4** is a plan view illustrating the stator core **23** in the embodiment. As illustrated in FIG. **4**, the stator core **23** is provided with an annular yoke part **31**, and the plurality of stator-core teeth parts **32-1** to **32-9** as drum parts, and is formed by stacking a plurality of metal plates. The metal plates are formed of, for example, soft magnetic bodies such as silicon steel plates.

(23) The yoke part **31** is formed approximately in a cylindrical shape. The first stator-core teeth part **32-1** among the plurality of stator-core teeth parts **32-1** to **32-9**, is formed approximately in a pillar shape. The first stator-core teeth part **32-1** is formed so that one end thereof is continuously formed in the inner peripheral side of the yoke part **31**, in other words, the first stator-core teeth part **32-1** is formed so as to project from the inner peripheral surface of the yoke part **31** toward the central axis of the stator core **23**. Similar to the first stator-core teeth part **32-1**, the stator-core teeth parts **32-2** to **32-9** other than the first stator-core teeth part **32-1** among the plurality of stator-core teeth parts **32-1** to **32-9**, are also formed in approximately pillar shapes and are projecting toward the central axis of the stator core **23** from the inner peripheral surface of the yoke part **31**. Also, the plurality of stator-core teeth parts **32-1** to **32-9** are formed on the inner peripheral surface of the yoke part **31** to be disposed at an equal interval of 40 degrees in the circumferential direction of the yoke part **31**. Hereinafter, the plurality of stator-core teeth parts **32-1** to **32-9** of the stator core **23** will be referred to as the stator-core teeth parts **32**.

(24) Also, as illustrated in FIG. **4**, on the outer peripheral surface of the stator core **23**, six first recessed parts **35** and three second recessed parts **36**, through which the refrigerant oil **8a** and the cooling medium pass, are provided, across the axial direction of the stator core **23**, at the positions corresponding to the stator-core teeth parts **32** in the circumferential direction of the stator core **23**. Two of the first recessed parts **35** are disposed to be interposed between the mutually adjacent

second recessed parts **36** in the circumferential direction of the stator core **23**. Also, a cutout groove **36a** for locating and attaching the upper insulator **24** and the lower insulator **25** with respect to the stator core **23**, is formed on each of the three second recessed parts **36**. A main part of the stator core **23** will be described later.

Characteristic Configurations of Compressor

(25) Next, characteristic configurations of the compressor **1** of the embodiment will be described. The characteristics of the embodiment include the stator core **23** in the three-phase motor **6**. In a manufacturing process of the compressor **1** of the present embodiment, the stator **22** is temporarily joined with the container **2** by shrink-fitting the container **2** onto the stator **22**. The stator **22** is temporarily joined in the container **2** and is then fixed by spot-welding onto the container **2**. The stator core **23** of the stator **22** in the embodiment is formed to have the dimensions that the outer diameter thereof implements transition-fitting with respect to the inner diameter of the container **2**, and the stator core **23** has the parts with different outer diameters in order to decrease the part which receives external force from the container **2** and to mitigate the external force applied from the container **2** when the container **2** is shrink-fitted onto the stator core **23**. As a result, the stator core **23** decreases the compression distortion, which is generated along with the external force applied from the container **2**.

(26) FIG. 5 is a plan view illustrating the main part of the stator core **23** in the embodiment. FIG. 6 is a plan view enlarging and illustrating a small-diameter part of the stator core **23** in the embodiment. As illustrated in FIG. 4, FIG. 5, and FIG. 6, in the stator core **23** of the stator **22**, in the circumferential direction of the stator core **23**, a plurality of large-diameter parts **37**, which are fitted on the inner peripheral surface of the container **2** by transition-fitting, and a plurality of small-diameter parts **38**, which are clearance-fitted on the inner peripheral surface of the container **2**, are formed to interpose therebetween the first recessed part **35** or the second recessed part **36** through which the refrigerant oil **8a** passes. In other words, in the circumferential direction of the stator core **23**, the large-diameter parts **37** and the small-diameter parts **38** are disposed at predetermined intervals therebetween. The large-diameter parts **37** and the small-diameter parts **38** mean parts of the outer peripheral surface of the annular yoke part **31** of the stator core **23**.

(27) Therefore, a relation of $R1 > R2$ is satisfied, wherein the distance from a center O of the stator core **23** serving as a rotation center of the three-phase motor **6** to the outer peripheral surface of the large-diameter part **37** (large-diameter-part outer diameter) is $R1$, and the distance from the center O to the outer peripheral surface of the small-diameter part **38** (small-diameter-part outer diameter) is $R2$. Also, a relation of $R3 > R2$ is satisfied, wherein the distance from the center O to the inner peripheral surface of the container **2** (housing inner diameter) is $R3$. Also, each of the small-diameter parts **38** clearance-fitted on the inner peripheral surface of the container **2** is formed so that a gap $G (=R3-R2)$ (see FIG. 6) between the outer peripheral surface of the small-diameter part **38** and the inner peripheral surface of the container **2**, is larger than 0 and equal to or lower than 250 [μm]. The gap G is set to, for example, $50 [\mu\text{m}] \leq G \leq 250 [\mu\text{m}]$.

(28) In other words, a contact part by which the stator core **23** contacts the inner peripheral surface of the container **2** when the container **2** is shrink-fitted onto the stator core **23**, is the large-diameter part **37**, and a non-contact part by which the stator core **23** does not contact the inner peripheral surface of the container **2** by clearance-fitting, is the small-diameter part **38**. The stator core **23**, which is fitted by shrink-fitting the large-diameter parts **37** in the container **2**, is spot-welded with the container **2** only by the small-diameter part **38**. Therefore, in the container **2**, the plurality of welded parts **20**, at which the inner peripheral surface of the container **2** and the outer peripheral surface of the stator core **23** are joined, is provided, and the welded parts **20** are provided only in the small-diameter parts **38**. Note that the large-diameter parts **37** are the contact parts for temporarily joining the stator core **23** in the container **2**, and at least two of the large-diameter parts **37** may be provided on the stator core **23** in order to implement the temporary joining.

(29) The stator core **23** in the present embodiment has the three large-diameter parts **37** and the six

small-diameter parts **38**. The three large-diameter parts **37** are disposed at an equal interval in the circumferential direction of the stator core **23**. The six small-diameter parts **38** are disposed so that two of the small-diameter parts **38** are disposed on each side to interpose each large-diameter part **37** in the circumferential direction of the stator core **23**. Note that the numbers of the large-diameter parts **37** and the small-diameter parts **38** are not limited to the above described numbers. Depending on the number of slots of the three-phase motor **6**, for example, the number of the small-diameter parts **38** is only needed to be two times or more of the number of the large-diameter parts **37**.

(30) In the circumferential direction of the stator core **23**, a circumferential-direction length $W1$ of the large-diameter part **37** along the outer peripheral surface of the stator core **23** and a circumferential-direction length $W2$ of the small-diameter part **38**, are formed to be equivalent to each other. Each large-diameter part **37** and each small-diameter part **38** are set to have a central angle of about 20 degrees about the center O of the stator core **23** (see FIG. 4).

(31) In the circumferential direction of the stator core **23**, a small-diameter range, which is a sum of the circumferential-direction lengths $W2$ of the plurality of small-diameter parts **38** along the outer peripheral surface of the stator core **23**, is larger than a large-diameter range, which is a sum of the circumferential-direction length $W1$ of the plurality of large-diameter parts **37**. In other words, $6W2$, which is the sum of the circumferential-direction lengths $W2$ of the six small-diameter parts **38**, is larger than $3W1$, which is the sum of the circumferential-direction lengths $W1$ of the three large-diameter parts **37**. As a result, in the circumferential direction of the stator core **23**, the large-diameter range, by which the outer peripheral surface of the stator core **23** contacts the inner peripheral surface of the container **2**, is decreased, and the external force, applied from the container **2** to the stator core **23** upon shrink-fitting, can be appropriately decreased.

(32) As illustrated in FIG. 4 and FIG. 5, the swaged part **28**, which integrates and fixes the plurality of stacked metal plates, is provided at each of the plurality of large-diameter parts **37**. The swaged parts **28** mutually join the metal plates by swaging so that part of the metal plates project in the thickness direction of the metal plates. The small-diameter parts **38** are not provided with the swaged parts **28**.

(33) FIG. 7 is a plan view enlarging and illustrating a through-hole of the stator core **23** in the embodiment. As illustrated in FIG. 5 and FIG. 7, each of the plurality of small-diameter parts **38** is provided with a through-hole **29**, which penetrates through the stator core **23** in the axial direction of the shaft **3**. The through-hole **29** is formed as a narrow long-hole-shaped opening, which extends in the circumferential direction of the stator core **23**. The through-hole **29** has arc-shaped long sides **29a**, which extend in the circumferential direction of the stator core **23**, and apexes **29b**, which are formed at both ends of the long sides **29a** in the circumferential direction of the stator core **23**.

(34) The through-hole **29** act as heat insulating space, which shields the heat applied to the stator core **23** upon welding of the small-diameter part **38** and the container **2**. Also, since part of the heat, which has reached the through-hole **29** from the welded part **20**, is caused to escape in the circumferential direction so as to be along the arc-shaped long side **29a**, the heat from the welded part **20**, does not easily travel around the through-hole **29**. Therefore, melting of the insulative film **26** (FIGS. 1 and 2), which is formed between the stator-core teeth parts **32**, due to the heat of welding, is suppressed. Note that from the viewpoint of shielding the heat which transmits from the welded part **20** to the stator core **23**, the position of the through-hole **29** with respect to the radial direction of the stator core **23**, is preferred to be disposed near the outer peripheral side of the small-diameter part **38**. Also, since each of the through-holes **29** is formed so that the center side thereof in the circumferential direction of the stator core **23** (vicinity of the midpoint of the long side **29a**) is the thickest, the heat from the part close to the welded part **20**, is not easily transmitted to the inner diameter side. In addition, since the through-hole **29** is provided with the arc-shaped long side **29a**, the flow of the magnetic flux, which pass through the stator core **23**, is not easily disturbed.

External Force and Stress Which Act on Stator Core in Manufacturing Processes of Compressor

(35) External force and stress on the stator core **23** configured in the above described manner will be described. FIG. **8** is a plan view for describing the external force and the stress which act on the stator core **23** in the manufacturing processes of the compressor **1** of the embodiment. FIG. **9** is a diagram illustrating, by analysis results of numerical analysis, the deformation, which is generated in the stator core **23** when the container **2** is shrink-fitted (temporarily joined) onto the stator core **23** in the manufacturing process of the compressor **1** of the embodiment. FIG. **10** is a diagram illustrating, by analysis results of numerical analysis, that the deformation of the stator core **23** generated by the shrink-fitting (temporary joining) in the manufacturing process of the compressor **1** of the embodiment is corrected (cancelled out) by welding after the temporary joining. In FIG. **10**, analysis has been carried out on the assumption of a state in which three small-diameter parts **38** among the six small-diameter parts **38** are welded with the container **2**.

(36) FIG. **8** is schematically illustrating the external force and the stress applied to the stator core **23** in the processes of the manufacturing processes of the compressor **1** of the embodiment. In the order of arrows, respectively, FIG. **8** illustrates the stator core **23** before the shrink-fitting, the stator core **23** temporarily joined with the container **2** by the large-diameter parts **37** at the three locations by the shrink-fitting, and the stator core **23** welded with the container **2** by the small-diameter parts **38** at the six locations after the temporary joining by the shrink-fitting.

(37) FIG. **9** illustrates analysis results of the deformation amount of the stator core **23** before and after the large-diameter parts **37** at the three locations are shrink-fitted onto the container **2** by the shrink-fitting illustrated in the first diagram and the second diagram of FIG. **8** in the manufacturing processes of the compressor **1** of the embodiment. In FIG. **9**, the analysis is carried out on the assumption that the external force by the shrink-fitting applied to each of the large-diameter parts **37** at the three locations, is about 6 MPa. In FIG. **9**, the line only with a contour illustrates the contour of the stator core **23** before the shrink-fitting, and the line with shading at the inside thereof illustrates the contour of the stator core **23** after the shrink-fitting. Note that in order to facilitate understanding of the manner of deformation, the deformation of the stator core **23** is illustrated in an exaggerated manner in FIG. **9**. Also, the location with thick shading represents the location with a large deformation amount before and after the shrink-fitting, and the location with pale shading represents the location with a small deformation amount before and after the shrink-fitting.

(38) FIG. **10** illustrates the analysis results of the deformation amount of the stator core **23** before the temporary joining by the shrink-fitting is carried out and after the small-diameter parts **38** at the six locations are welded onto the container **2** by the welding after the shrink-fitting illustrated in the first diagram and the third diagram of FIG. **8** in the manufacturing processes of the compressor **1** of the embodiment. In FIG. **10**, the analysis has been carried out on the assumption that the external force caused by the shrink-fitting and applied to each of the large-diameter parts **37** at the three locations, is about 6 MPa, and on the assumption that the external force caused by the welding and applied to each of the small-diameter parts **38** at the six locations, is 5 to 20 MPa. In FIG. **10**, the line only with a contour illustrates the contour of the stator core **23** illustrated in the first diagram of FIG. **8** before the shrink-fitting, and the line with shading at the inside thereof illustrates the contour of the stator core **23** illustrated in the third diagram of FIG. **8** after the shrink-fitting and the welding is carried out. Similar to FIG. **9**, in order to facilitate understanding of the manner of deformation, the deformation of the stator core **23** is illustrated in an exaggerated manner in FIG. **10**. Also, the location with thick shading represents the location with a large deformation amount before and after the shrink-fitting and the welding is carried out, and the location with pale shading represents the location with a small deformation amount before and after the shrink-fitting and the welding is carried out.

(39) As illustrated in FIG. **8** and FIG. **9**, in the manufacturing processes of the compressor **1**, the container **2** is shrink-fitted onto the stator core **23** in a state that the outer peripheral surfaces of the large-diameter parts **37** of the stator core **23** are fitted onto the inner peripheral surface of the container **2** by transition-fitting and that the outer peripheral surfaces of the small-diameter parts **38**

are fitted onto the inner peripheral surface of the container **2** by clearance-fitting. As a result, the container **2** and the stator core **23** are temporarily joined by the large-diameter parts **37**.

(40) In a case of the three-phase motor **6** with nine slots, when a conventional stator core is temporarily joined in the container **2**, nine locations in the circumferential direction of the stator core contact the interior of the container **2** (nine-face fitting). Compared with this structure, in the embodiment, only the three large-diameter parts **37** at the three locations in the circumferential direction of the stator core **23**, contact the interior of the container **2** (three-face fitting). Therefore, in the embodiment, the locations by which the stator core **23** receives external force from the container **2**, which is shrink-fitted on the stator core **23**, can be decreased from the nine locations to the three locations. Therefore, in the embodiment, the external force applied from the container **2** to the stator core **23**, is decreased.

(41) Also, even in a case in which interference is generated with respect to the container **2** ($R1 > R3$) when the large-diameter parts **37**, which are transition-fitted onto the container **2**, have a lower limit of dimension tolerance due to dimensional variations of the stator core **23**, the locations by which the stator core **23** receives the external force from the welded parts **20**, can be decreased, since only the three large-diameter parts **37** contact the interior of the container **2**.

(42) When the container **2** is shrink-fitted onto the stator core **23** in the above described manner, as illustrated in FIG. **8** and FIG. **9**, the stator core **23** receives, by the three large-diameter parts **37** contacting the inner peripheral surface of the container **2**, the external force toward the center O of the stator core **23** from the container **2** (see solid-line arrows in the center diagram of FIG. **8**) in the plane orthogonal to the central axis of the stator core **23**. Also, as the three large-diameter part **37** receive the external force toward the center O, in the interior of the stator core **23**, stress (compression stress) toward the outer peripheral side is generated, as resistance force against the external force from the welded part **20**, in the center part between the large-diameter parts **37** which are adjacent to each other in the circumferential direction of the stator core **23** (see broken-line arrows in the center diagram of FIG. **8**). Therefore, as understood from the differences between the lines only with contours and the lines with shading at the inside thereof in FIG. **9**, because of the shrink-fitting, the stator core **23** is deformed into a triangular shape, in which vicinities of the three large-diameter parts **37** are crushed toward the inner diameter side and vicinities of the six small-diameter parts **38** are pushed outward toward the outer diameter side.

(43) Subsequently, as illustrated in FIG. **8** and FIG. **10**, the welded parts **20** are formed since the small-diameter parts **38** of the stator core **23** are spot-welded with the container **2**, after the stator core **23** is shrink-fitted in the container **2**. When the small-diameter parts **38** and the container **2** are joined by the welded parts **20**, as illustrated in FIG. **8**, the six small-diameter parts **38** receive external force from the welded parts **20** in the direction toward the center O of the stator core **23**. At the same time, as the small-diameter parts **38** receive the external force toward the center O, stress (compression stress) toward the outer peripheral side of the stator core **23** is generated in the large-diameter parts **37** as the resistance force against the external force. Since the external force and the stress like this act on the small-diameter parts **38** and the large-diameter parts **37**, the deformation of the triangular shape generated in the stator core **23** along with the shrink-fitting of the container **2**, is corrected to a circular shape on the plane orthogonal to the central axis of the stator core **23**. By virtue of this, the shape of the stator core **23** in the inner peripheral side can be maintained in a circular shape along the outer peripheral surface of the rotor **21**, and inequality of air gaps between the stator core **23** and the rotor **21**, can be decreased. Therefore, magnetic balance can be equalized in the circumferential direction of the stator core **23**, deterioration of vibrations can be prevented, and noise can be suppressed. Also, since the shape of the stator core **23** in the inner peripheral side is maintained in the circular shape, insertion of the rotor **21** in the inner peripheral side of the stator core **23** can be facilitated, and deterioration in the assemblability of the three-phase motor **6** due to dimensional variations, can be prevented.

(44) Herein, a condition for generation of an interference S between the large-diameter part **37** of

the stator core **23** and the container **2**, is that the large-diameter-part outer diameter **R1** of the large-diameter part **37** of the stator core **23** and the housing inner diameter **R3** of the container **2** satisfy the relation of $R1 > R3$ at the point before the shrink-fitting, and the interference **S** herein can be expressed by $S = R1 - R3$. Therefore, when the interference **S** is generated, $R1 > R3$ is satisfied according to $S = R1 - R3 > 0$. On the other hand, a condition for generating the gap **G** between the small-diameter part **38** of the stator core **23** and the container **2**, is that the small-diameter-part outer diameter **R2** of the small-diameter part **38** of the stator core **23** and the housing inner diameter **R3** of the container **2** satisfy the relation of $R3 > R2$ at the point before the shrink-fitting, and the gap **G** herein is expressed by $G = R3 - R2$. Therefore, when the gap **G** is generated, $R3 > R2$ is satisfied according to $G = R3 - R2 > 0$. With these two conditions together, when $R2 < R3 < R1$ is satisfied at the point before the shrink-fitting, the deformation generated due to the shrink-fitting can be mitigated by the welding.

(45) When the small-diameter parts **38** of the deformed stator core **23** are welded with the container **2**, the external force toward the center **O** of the stator core **23** acts on the small-diameter parts **38** as described above. Therefore, load (referred to as welding load.) is applied to the welded parts **20** of the stator core **23**. Depending on the welding load applied to the welded parts **20**, the inner diameter of the teeth parts **32** of the stator core **23** from the center **O**, is changed.

Correction of Deformation of Stator Core by Welding of Small-Diameter Parts

(46) FIG. **11** is a graph illustrating that the deformation generated in the stator core **23** along with the shrink-fitting of the container **2** in the above described manner, is corrected by welding the small-diameter parts **38** with the container **2**. FIG. **11** illustrates the deformation amounts of the locations, at which the teeth parts **32** of the stator core **23** have been deformed the most by the shrink-fitting or welding, in the case, in which the interference **S** (the difference between the large-diameter-part outer diameter **R1** of the large-diameter part **37** of the stator core **23** and the housing inner diameter **R3** of the container **2**: $R1 - R3$) of the stator core **23** is equal to or higher than 0 in transition-fitting. In FIG. **11**, the horizontal axis illustrates the interference **S** [μm] of the stator core **23**, and the vertical axis illustrates maximum displacement amounts [μm] of the inner diameters of the teeth parts **32** of the stator core **23**.

(47) In FIG. **11**, **Q1** represents the maximum deformation amounts generated in the teeth parts **32** of the stator core **23** when the container **2** is shrink-fitted by the large-diameter parts **37** at the three locations of the stator core **23** in the manufacturing processes of the compressor **1** of the embodiment. As illustrated by **Q1** in FIG. **11**, it can be understood that, in the case only with the shrink-fitting by the three large-diameter parts **37** of the stator core **23** (at the point before the welding of the six small-diameter parts **38**), the larger the interference **S** of the stator core **23**, the larger the deformation of the teeth parts **32** of the stator core **23**.

(48) In FIGS. **11**, **Q2** to **Q5** represent the maximum deformation amounts generated in the teeth parts **32** of the stator core **23** when the stator core **23** after the shrink-fitting by the large-diameter parts **37** at the three locations are further welded by the small-diameter parts **38** at the six locations of the stator core **23** in the manufacturing processes of the compressor **1** of the embodiment. In each of **Q2** to **Q5**, analysis has been carried out on the assumption that the external force expected to be applied to the stator core **23** from each of the welded parts **20** at the six locations upon welding is 5 to 20 MPa at an interval of 5 MPa. Herein, the inner diameter of the stator core **23** means an inner diameter **R4** of the teeth part **32** (see FIGS. **4** and **5**).

(49) First, focusing on **Q2** of FIG. **11** (welding load: 5 MPa), when the interference of the stator core **23** is 0 μm , the displacement amount of the inner diameter **R4** of the stator core **23** is about 6 μm , and the displacement amount is larger than 0 μm of the case of **Q1** only with the shrink-fitting (without welding). In other words, when the interference is 0 μm , the stator core **23** is largely deformed by the welding more than before the welding. On the other hand, the displacement amount of the stator core **23** is about 5 μm when the interference of the stator core **23** of **Q2** is 20 μm , and the displacement amount is smaller than the displacement amount 8 μm of the stator core

23 in the case in which the interference of **Q1** is 20 μm . In other words, when the welding load is 5 MPa and the interference is 20 μm , the deformation of the stator core **23** can be mitigated by the welding more than before the welding. Similarly, in the range in which the interference of the stator core **23** is 40 μm or higher, when the cases of the same interference are mutually compared, the displacement amount of the inner diameter **R4** of the stator core **23** (deformation amount of the stator core **23**) is mitigated after the welding more than before the welding. Therefore, it can be said that, in the case in which the welding load is 5 MPa, the deformation of the stator core **23** is mitigated after the welding more than before the welding in the range in which the interference is equal to or higher than 20 μm (larger than 10 μm).

(50) Similarly, focusing on **Q3** of FIG. 11 (welding load: 10 MPa), when the interference of the stator core **23** is 0 μm , the displacement amount of the inner diameter **R4** of the stator core **23** is about 13 μm , and the displacement amount is larger than 0 μm of the case of **Q1** only with the shrink-fitting (without welding). In other words, also in **Q3**, when the interference is 0 μm , the stator core **23** is largely deformed by the welding more than before the welding. On the other hand, when the interference of the stator core **23** is 20 μm , the displacement amount of the inner diameter **R4** of the stator core **23** of **Q3** is about 8 μm , and the deformation amount of the stator core **23** is not changed compared with the case of **Q1** before the welding. On the other hand, when the interference of the stator core **23** is 40 μm , the displacement amount of the stator core **23** of **Q3** is about 9 μm , and the displacement amount is smaller than the displacement amount 16 μm of the stator core **23** in the case in which the interference of **Q1** is 40 μm . In other words, when the welding load is 10 MPa and the interference is 40 μm , the deformation of the stator core **23** can be mitigated by the welding more than before the welding. Similarly, in the range in which the interference of the stator core **23** is 60 μm or higher, when the cases of the same interference are mutually compared, the displacement amount of the inner diameter **R4** of the stator core **23** (deformation amount of the stator core **23**) is mitigated after the welding more than before the welding. Therefore, it can be said that, in the case in which the welding load is 10 MPa, the deformation of the stator core **23** is mitigated after the welding more than before the welding in the range in which the interference is larger than 20 μm .

(51) Similarly, it can be said that, according to **Q4** (welding load: 15 MPa) of FIG. 11, in the case in which the welding load is 15 MPa, the deformation of the stator core **23** is mitigated after the welding more than before the welding in the range in which the interference is equal to or higher than 40 μm (larger than 30 μm). Similarly, it can be said that, according to **Q5** (welding load: 20 MPa) of FIG. 11, in the case in which the welding load is 20 MPa, the deformation of the stator core **23** is mitigated after the welding more than before the welding in the range in which the interference is larger than 40 μm .

(52) In other words, as illustrated in FIG. 11, at the point before the welding which is illustrated by **Q1**, the larger the interference of the stator core **23** (the larger the external force received by the stator core **23** from the container **2**) in the case in which the shrink-fitting is carried out, the larger the displacement amount of the inner diameter **R4** of the stator core **23**. On the other hand, at the points after the welding illustrated by **Q2** to **Q5**, when the line of **Q1** before the welding is used as a reference, in the combinations of the interference and the welding load which are positioned below **Q1**, the displacement amount of the inner diameter **R4** of the stator core **23** is increased (in other words, the stator core **23** is largely deformed by the welding more than before the welding). However, in the combinations of the interference and the welding load of **Q2** to **Q5** which are positioned above the line of **Q1**, which is before the welding, the displacement amount of the inner diameter **R4** of the stator core **23** is low and suppressed after the welding more than the point before the welding (in other words, deformation of the stator core **23** is suppressed by carrying out the welding more than before the welding). Herein, according to FIG. 11, it can be understood that the deformation is mitigated after the welding within the range that satisfies $S/Q > 20/10 = 2$ [$\mu\text{m}/\text{MPa}$], wherein the welding load is Q (MPa), and the interference in the shrink-fitting is S

(μm).

(53) As described above, the deformation generated in the stator core **23** by the shrink-fitting, is corrected by the welded parts **20** by which the small-diameter parts **38** are joined with the container **2**. Also in the case in which the interference of the stator core **23** increases, increase in the displacement amount of the stator core **23** is suppressed by the welded parts **20**. Therefore, the welded parts **20** by which the small-diameter parts **38** are joined with the container **2**, can suppress the deformation of the stator core **23** and mitigate the distortion of the air gaps between the stator core **23** and the rotor **21**.

Comparison between Embodiment and Comparative Example

(54) FIG. **12** is a graph illustrating the relation between the interference of the stator core **23** in the embodiment and the comparative example and decrease rates of motor efficiency (the ratio of mechanical energy output from the three-phase motor **6** with respect to the electric energy input to the three-phase motor **6**). In FIG. **12**, the vertical axis illustrates the decrease rate [%] of the motor efficiency, and the horizontal axis illustrates the interference [μm] of the stator core **23**. The comparative example has a nine-face fitting structure in which nine locations on the outer peripheral surface of the stator core **23** contact the inner peripheral surface of the container **2**, and the decrease rates of the motor efficiency of the comparative example are illustrated by a solid line **L1** and a broken line **L2** in FIG. **12**. The embodiment has the three-face fitting structure in which the outer peripheral surface of the three large-diameter parts **37** of the stator core **23** contact the inner peripheral surface of the container **2**, and the decrease rates of the motor efficiency of the embodiment are illustrated by a solid line **L3** and a broken line **L4** in FIG. **12**. In FIG. **12**, the solid lines **L1** and **L3** are analytical values, and the broken lines **L2** and **L4** are actually measured values. Also, a broken line **M** is a maximum interference which is a lower limit of fitting tolerance and is, for example, set to about 60 to 80 [μm].

(55) As illustrated in FIG. **12**, the motor efficiency is lowered as the interference of the stator core increases in both of the embodiment and the comparative example. However, the decrease rates of the motor efficiency in the embodiment illustrated by the solid line **L3** and the broken line **L4** are smaller than the decrease rates of the motor efficiency in the comparative example illustrated by the solid line **L1** and the broken line **L2**. In other words, according to the three-face fitting of the embodiment, decrease in the motor efficiency can be suppressed more than the nine-face fitting structure of the comparative example.

(56) FIG. **13** is a graph for describing APF efficiency (annual performance factor) in the embodiment. FIG. **13** illustrates changes in the APF efficiency before the shrink-fitting and after the shrink-fitting of the container **2** with respect to the stator core regarding an embodiment and a comparative example. Similar to the above description, the comparative example has the nine-face fitting structure. The embodiment has the three-face fitting structure.

(57) As illustrated in FIG. **13**, in the comparative example, the APF efficiency is 91.58% before the shrink-fitting of the container **2**. However, the APF efficiency is lowered to 91.19 [%] after the shrink-fitting of the container **2**. On the other hand, in the embodiment, the APF efficiency is 91.58% before the shrink-fitting of the container **2**. However, the APF efficiency is 91.44 [%] after the shrink-fitting of the container **2**. Thus, in the embodiment, decrease in the APF efficiency before and after the shrink-fitting, is suppressed more than the comparative example. This is conceivably for a reason that, in the embodiment provided with the large-diameter parts **37** on three faces and the small-diameter parts **38** on six faces, external force is no longer applied on the small-diameter parts **38**, and distortion of the stator core **23** is suppressed while, in the comparative example provided with the large-diameter parts on nine faces, external force is applied to all of the large-diameter parts.

Effects of Embodiment

(58) As described above, in the three-phase motor **6** of the compressor **1** of the embodiment, in the stator core **23** of the stator **22**, the plurality of large-diameter parts **37**, which contact the inner

peripheral surface of the container 2 when fitted by the shrink-fitting, and the plurality of small-diameter parts 38, which has the distance from the center O to the outer peripheral surface smaller than the distance of the large-diameter part 37, are formed so that the first recessed part 35 or the second recessed part 36 is interposed therebetween in the circumferential direction of the stator core 23. The container 2 and the stator core 23 are provided with the plurality of welded parts 20 by which the inner peripheral surface of the container 2 and the outer peripheral surface of the stator core 23 are joined, and the welded parts 20 are provided only in the small-diameter parts 38 in the stator core 23. Moreover, the small-diameter parts 38 do not contact the inner peripheral surface of the container 2 except for the welded parts 20 since the small-diameter parts are clearance-fitted on the inner peripheral surface of the container 2. By virtue of this, assemblability of the compressor 1 can be ensured sincebe temporarily joined on the inner peripheral surface of the container 2 when the large-diameter parts 37 are fitted by shrink-fitting on the inner peripheral surface of the container 2. Moreover, since the small-diameter parts 38 are clearance-fitted on the inner peripheral surface of the container 2, the area by which the outer peripheral surface of the stator core 23 contacts the inner peripheral surface of the container 2 in the circumferential direction of the stator core 23 can be decreased. Therefore, compression distortion generated in the stator core 23 can be decreased. Moreover, since the welded parts 20 by which the small-diameter parts 38 are joined with the container 2 are provided, the deformation generated in the stator core 23 upon the shrink-fitting can be corrected while firmly fixing the stator core 23 to the container 2, and the air gaps between the stator core 23 and the rotor 21 can be uniformized in the circumferential direction. Therefore, according to the embodiment, decrease in the efficiency of the three-phase motor 6 can be suppressed by decreasing the distortion of the stator core 23 while ensuring the assemblability of the compressor 1. Moreover, as illustrated in FIG. 11, since the small-diameter parts 38 are welded, displacement of the inner diameter R4 of the stator core 23 (deformation of the stator core 23) after the shrink-fitting can be mitigated more than the point before the welding is carried out.

Furthermore, as illustrated in FIG. 12, the decrease in the motor efficiency caused along with the size of the interference of the shrink-fitting, can be suppressed. Moreover, as illustrated in FIG. 13, since the temporary joining is carried out with the large-diameter parts 37 of the three faces with respect to the container 2, the decrease in the APF efficiency before and after the shrink-fitting can be suppressed more than the case in which temporary joining is carried out with all the nine faces with respect to the container 2.

(59) Moreover, in the compressor 1 of the embodiment, in the circumferential direction of the stator core 23, a small-diameter range, which is a sum of the circumferential-direction lengths W2 of the plurality of small-diameter parts 38 along the outer peripheral surface of the stator core 23, is larger than a large-diameter range, which is a sum of the circumferential-direction length W1 of the plurality of large-diameter parts 37. By virtue of this, in the circumferential direction of the stator core 23, the large-diameter range by which the outer peripheral surface of the stator core 23 contacts the inner peripheral surface of the container 2 is decreased, and the external force and stress applied to the stator core 23 can be decreased.

(60) Moreover, in the compressor 1 of the embodiment, each of the plurality of small-diameter parts 38 of the stator core 23 is provided with the through-hole 29, which penetrates through the stator core 23 along the axial direction of the shaft 3. By virtue of this, the through-holes 29 act as heat insulating spaces, which insulate the heat applied to the stator core 23 upon the welding of the small-diameter parts 38 and the container 2. Therefore, the insulative film 26 of the slots can be prevented from being melted by the heat in the welding.

(61) Moreover, in the compressor 1 of the embodiment, the through-hole 29 of the small-diameter part 38 of the stator core 23 is formed to extend in the circumferential direction of the stator core 23, and the length of the through-hole 29 in the circumferential direction of the stator core 23 is formed to be larger than the length of the welded part 20 in the circumferential direction of the stator core 23. By virtue of this, heat is caused to escape along the circumferential direction of the

stator core **23** while the heat transmitted from the welded part **20** is shielded by the through-hole **29**. Therefore, transmission of the heat toward the inner diameter side of the stator core **23** around the through-hole **29**, can be suppressed. Therefore, melting of the insulative film **26** due to the heat in the welding, can be further prevented.

(62) Note that the compressor **1** of the present embodiment is not limited to a rotary compressor, but may be applied to another compressor such as a scroll compressor.

REFERENCE SIGNS LIST

(63) **1** COMPRESSOR **2** CONTAINER (COMPRESSOR HOUSING) **3** SHAFT (ROTARY SHAFT) **5** COMPRESSION PART **6** THREE-PHASE MOTOR **8a** REFRIGERANT OIL (LUBRICATING OIL) **11** INTAKE TUBE (INTAKE PART) **12** DISCHARGE TUBE (DISCHARGE PART) **20** WELDED PART **21** ROTOR **22** STATOR **23** STATOR CORE **29** THROUGH-HOLE **35** FIRST RECESSED PART (RECESSED PART) **36** SECOND RECESSED PART (RECESSED PART) **37** LARGE-DIAMETER PART **38** SMALL-DIAMETER PART **G** GAP **O** ROTATION CENTER (CENTER) **R1** LARGE-DIAMETER-PART OUTER DIAMETER **R2** SMALL-DIAMETER-PART OUTER DIAMETER **R3** HOUSING INNER DIAMETER **R4** INNER DIAMETER OF STATOR CORE **S** INTERFERENCE **W1, W2** CIRCUMFERENTIAL-DIRECTION LENGTH

Claims

1. A compressor comprising: a compressor housing; a compression part that is disposed in the compressor housing and compresses a cooling medium; and a motor that is disposed in the compressor housing and drives the compression part, wherein the motor has a rotor provided coaxially with respect to a rotary shaft of the compression part, and a stator core disposed on an outer peripheral side of the rotor, the stator core has formed thereon, in a circumferential direction, a plurality of large-diameter parts shrink-fitted on an inner peripheral surface of the compressor housing, a plurality of small-diameter parts each having a distance from a rotation center of the motor to an outer peripheral surface of the stator core that is smaller than the distance of the large-diameter part, and a plurality of recessed parts, which is formed at an interval in a circumferential direction of the outer peripheral surface of the stator core and through which lubricating oil in the compressor housing passes along the outer peripheral surface, the small-diameter part and the large-diameter part adjacent to each other in the circumferential direction of the stator core are formed to interpose the recessed part between the small-diameter part and the large-diameter part, and the stator core is provided with a plurality of welded parts joining the inner peripheral surface of the compressor housing with the outer peripheral surface of the stator core, and the welded parts are provided in the small-diameter parts of the stator core.
2. The compressor according to claim 1, wherein, in the circumferential direction of the stator core, a circumferential-direction length of the large-diameter part along the outer peripheral surface of the stator core and a circumferential-direction length of the small-diameter part are equal to each other.
3. The compressor according to claim 1, wherein, in the circumferential direction of the stator core, a small-diameter range, which is a sum of the circumferential-direction lengths of the plurality of small-diameter parts along the outer peripheral surface of the stator core, is larger than a large-diameter range, which is a sum of the circumferential-direction lengths of the plurality of large-diameter parts.
4. The compressor according to claim 1, wherein, in the stator core, a number of the small-diameter parts is two times or more a number of the large-diameter parts.
5. The compressor according to claim 1, wherein each of the plurality of small-diameter parts is provided with a through-hole that penetrates through the stator core along an axial direction of the rotary shaft.

6. A compressor comprising: a compressor housing; a compression part that is disposed in the compressor housing and compresses a cooling medium; and a motor that is disposed in the compressor housing and drives the compression part, wherein the motor has a rotor provided coaxially with respect to a rotary shaft of the compression part, and a stator core disposed on an outer peripheral side of the rotor, the stator core has formed thereon, in a circumferential direction, a plurality of large-diameter parts in contact with an inner peripheral surface of the compressor housing and a plurality of small-diameter parts each having a distance from a rotation center of the motor to an outer peripheral surface that is smaller than the distance of the large-diameter part, the stator core is provided with a plurality of welded parts joining the inner peripheral surface of the compressor housing with the outer peripheral surface of the stator core, and the welded parts are provided in the small-diameter parts of the stator core, and before arranging the stator core in the compressor housing, $R2 < R3 < R1$ is satisfied, wherein $R1$ represents a large-diameter-part outer diameter that is a distance from the rotation center to the outer peripheral surface of the large-diameter part, $R2$ represents a small-diameter-part outer diameter that is a distance from the rotation center to the outer peripheral surface of the small-diameter part, and $R3$ represents a housing inner diameter that is a distance from the rotation center to the inner peripheral surface of the compressor housing, and the stator core is shrink-fitted onto the compressor housing by the large-diameter part and is welded with the compressor housing by the small-diameter part.
